

Thomas Marconi

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science

Atlanta, GA

Expected Dec 2027

- Coursework: DB System Implementation, HCI, High-Performance Computer Architecture, Computer Networks

Florida State University

B.S. Computer Science | GPA: 3.7

Tallahassee, FL

May 2023

EXPERIENCE

Tactical Air Support | Aerospace defense contractor supporting U.S. military ops

Feb 2024 – Present

Senior Software Engineer

Jacksonville, FL

- Designed a real-time collaborative task management system requiring concurrent write coordination across connected clients; solved event ordering and partial-failure recovery at the WebSocket layer, achieving consistent sub-100ms state propagation without an external message broker.
- Replaced three vendor software systems with purpose-built internal tooling, eliminating **\$850K in annual licensing costs**; engineered for high availability and zero-downtime deployments with no dedicated operations team.
- Architected runtime secrets injection across a multi-service Azure deployment using managed identities and Key Vault; defined least-privilege IAM policies for all service principals and eliminated static credentials from the codebase to satisfy FedRAMP High audit controls.
- Built a document retrieval pipeline over 5,000+ unstructured PDFs; designed context-aware chunking with overlapping windows to preserve semantic coherence across boundaries, and implemented two-stage retrieval (dense vector similarity followed by sparse reranking) to improve precision on narrow technical queries.

Florida State University

Aug 2022 – Dec 2022

Teaching Assistant — Data Structures & Algorithms

Tallahassee, FL

- Conducted code reviews for 30+ students on C++ implementations of trees, graphs, and hash tables; focused on heap allocation correctness, pointer semantics, and worst-case complexity tradeoffs.

TECHNICAL PROJECTS

Project Desert | Next.js, Python (FastAPI), PostgreSQL, Docker

Jan 2025 – Present

- Enforced explicit API contracts via OpenAPI schema validation at service boundaries; auto-generated TypeScript clients from Pydantic models to surface type violations at compile time and eliminate a class of cross-service integration failures.
- Implemented stateless JWT authentication with server-side refresh token rotation across distributed services; engineered to handle concurrent refresh races and session invalidation without shared distributed state.

CityScale | React, FastAPI, PCA | 3rd Place — AI for Good Hackathon

Oct 2025

- Built a geospatial rendering pipeline capable of filtering and visualizing **580,000+ data points** at sub-second latency; implemented client-side spatial indexing and progressive data loading to maintain interactive frame rates at full dataset scale.
- Applied PCA to a high-dimensional economic dataset; selected retained components by explained variance ratio to produce a compact 2D city-level embedding while preserving the majority of discriminative signal.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, C++, Java, Bash

Backend: Node.js, FastAPI, WebSockets, REST APIs, PostgreSQL, Docker, Podman

AI & Data: Vector Retrieval (ChromaDB, Azure Semantic Search), LLM Orchestration (LangGraph), Pandas, scikit-learn

Cloud & Infrastructure: Azure (Functions, Managed Identity, Key Vault, Entra ID), CI/CD, Linux

Frontend: React, Next.js, Zustand, TanStack Query, Tailwind CSS